Case study 2

Name - Tanay Chetan Mehta

Roll No. - 23BME010

Topic - Nonlinear Dynamic Modelling and Machine Learning–Based Power Prediction of a Tidal Current Turbine System

Problem Description

Tidal current turbines are used to harness the kinetic energy of moving seawater and convert it into useful power. In this project, the turbine system is modelled as a rotating mechanical setup influenced by hydrodynamic forces generated due to tidal currents. The system considers key physical parameters such as moment of inertia, damping, shaft stiffness, seawater density, turbine radius, and torque coefficient.

The tidal velocity is not constant and varies with time due to natural tidal cycles. To represent this behaviour, it is modelled as a sinusoidal function. The objective of the study is to analyze how the turbine responds to this time-varying input in terms of angular displacement and velocity, and to determine the corresponding power output. Additionally, a machine learning model is used to predict power output based on tidal velocity data, making the system useful for both simulation and prediction purposes.

Governing Equations

The behaviour of the tidal turbine system is governed by fundamental principles of rotational dynamics and fluid mechanics. The rotational form of Newton's Second Law is used to describe the motion of the turbine shaft, where the applied hydrodynamic torque drives the system against inertia, damping, and stiffness effects.

$$\mathrm{I}\frac{d^2\theta}{dt^2} + \mathrm{c}\frac{d\theta}{dt} + \mathrm{k}\theta = T_{fluid}$$

The hydrodynamic torque acting on the turbine blades is calculated based on the properties of the fluid and turbine geometry. It depends on the seawater density, swept area of the turbine, radius, torque coefficient, and the square of the tidal velocity.

$$T_{fluid} = \frac{1}{2}\rho A R C_t v^2$$

The instantaneous power output of the turbine is determined by multiplying the torque with the angular velocity of the shaft.

$$P(t) = T_{fluid}(t) \cdot \dot{\theta}(t)$$

These equations together form the foundation of the dynamic model used in the project.

Working Principle

The working principle of a tidal current turbine is similar to that of a wind turbine, but it operates in water instead of air. As tidal currents flow, they exert a force on the turbine blades, generating hydrodynamic torque. This torque causes the turbine shaft to rotate.

The rotational motion is influenced by the system's inertia, damping, and stiffness, which control how the turbine accelerates and stabilizes over time. As the shaft rotates, it develops angular velocity, and the interaction between torque and angular velocity produces mechanical power.

Since tidal velocity varies continuously with time, the power output also changes accordingly. Higher velocities result in greater torque and increased power generation. The system dynamically adjusts to these changes, and the resulting data is used both for analysis and for training a machine learning model to predict power output efficiently.

Mathematical Model

The tidal turbine system is modelled as a second-order nonlinear dynamic system that represents the rotational motion of the turbine shaft under the influence of hydrodynamic forces. The primary variable used to describe the system is the angular displacement $\left(\theta(t)\right)$, while angular velocity and angular acceleration are derived from it.

The fundamental equation governing the system is based on the rotational form of Newton's Second Law. It accounts for inertia, damping, and stiffness effects acting on the turbine shaft:

$$I\frac{d^2\theta}{dt^2} + c\frac{d\theta}{dt} + k\theta = T_{fluid}$$

The input to this system is the tidal velocity, which varies with time and is modelled as a sinusoidal function to represent real tidal behaviour:

$$v(t) = V_0 \sin(\omega t)$$

The hydrodynamic torque acting on the turbine blades depends on this velocity and is given by:

$$T_{fluid} = \frac{1}{2}\rho A R C_t (V_0 \sin(\omega t))^2$$

Substituting the torque expression into the governing equation gives the complete nonlinear dynamic equation of the system:

$$I\frac{d^2\theta}{dt^2} + c\frac{d\theta}{dt} + k\theta = \frac{1}{2}\rho A R C_t (V_0 \sin(\omega t))^2$$

This equation is solved numerically using MATLAB to obtain angular displacement and angular velocity as functions of time.

Once the angular velocity is known, the instantaneous power generated by the turbine is calculated as:

$$P(t) = T_{fluid}(t) \cdot \dot{\theta}(t)$$

This mathematical formulation captures the interaction between fluid flow and mechanical response, allowing accurate prediction of turbine behaviour under varying tidal conditions.

Assumptions Taken for MATLAB modelling

- Tidal flow is assumed to be uniform and steady across the turbine blades, neglecting turbulence and wake effects.
- The torque coefficient (C_t) is considered constant, ignoring its variation with operating conditions.
- The turbine shaft is modelled as a linear spring-damper system for simplicity.
- Blade deformation, material fatigue, and nonlinear stiffness effects are neglected.
- Energy losses other than damping (such as electrical losses) are not considered.
- These assumptions simplify the model while maintaining reasonable accuracy for analysis.

Brief Description of Machine Learning Used in the Project

In addition to the physics-based model, a machine learning approach is used to predict power output from tidal velocity data. A polynomial regression model is trained using the simulated dataset generated from MATLAB. This model establishes a mathematical relationship between velocity and power without explicitly solving the governing equations.

The purpose of incorporating machine learning is to provide a fast and efficient way to estimate power output once sufficient data is available. Instead of repeatedly solving differential equations, the trained model can directly predict power for given velocity inputs. This approach is particularly useful for real-time applications and optimization studies.

Applications, Optimisation, and Scope

Tidal turbine systems are primarily used for renewable energy generation in coastal and offshore regions with strong tidal currents. Countries with predictable tidal patterns can use this technology to supplement their energy needs in a sustainable way.

From an optimization perspective, the model can be used to study how design parameters such as turbine radius, damping, and stiffness affect performance. It also helps in selecting optimal operating conditions to maximize power output while minimizing mechanical stress.

The scope of this project extends to hybrid energy systems, smart grids, and predictive maintenance, where machine learning models can assist in decision-making and operational efficiency.

Countries Currently Using This Technology

Tidal energy technology is being actively explored and implemented in countries with strong tidal resources. The United Kingdom is one of the global leaders, with projects in Scotland such as the MeyGen tidal array. France has long utilized tidal energy through the La Rance tidal power station.

Other countries like Canada, South Korea, and China are also investing in this technology. India has potential in regions like the Gulf of Khambhat but is still in the developmental stage.

Effects of Various Parameters on Power Output

Power output is highly sensitive to tidal velocity, as it varies with the square of velocity. Even small increases in current speed can significantly boost power generation. The turbine radius and swept area also play a major role, as larger blades capture more energy from the flow.

Damping affects how quickly the system stabilizes, while stiffness influences oscillatory behaviour. High damping reduces fluctuations but may lower peak performance, whereas low damping can lead to instability.

Weather conditions, such as storms and rough sea states, can disrupt normal operation. During extreme tidal waves, turbines may be shut down or operated in standby mode to prevent structural damage. This directly reduces power output but is necessary for safety and longevity of the system.

Problems and Challenges in Implementation

Implementing tidal turbine systems comes with several practical challenges. Marine environments are harsh, leading to issues like corrosion, biofouling, and mechanical wear. Installation and maintenance in underwater conditions are complex and expensive.

Another major challenge is variability in tidal currents, which affects consistent power generation. Grid integration can also be difficult due to fluctuating output. Additionally, environmental concerns such as impact on marine life need to be carefully addressed.

Power Production Outcome and Scaling

The simulation results show that power output follows the trend of tidal velocity, increasing during peak flow and decreasing during low flow periods. The system reaches a stable operating condition after initial transients.

Scaling the project involves increasing turbine size, deploying multiple turbines in arrays, or optimizing placement in high-velocity regions. However, scaling must consider structural limits, environmental impact, and economic feasibility.

MATLAB Code Description

The MATLAB code developed for this project simulates the dynamic behavior of a tidal current turbine and calculates its power output under time-varying tidal conditions. The model is implemented using a numerical approach to solve the governing nonlinear differential equation of motion.

The code begins by defining all the necessary physical parameters, including seawater density, turbine radius, swept area, torque coefficient, moment of inertia, damping coefficient, and shaft stiffness. These parameters represent the physical characteristics of the turbine system and the surrounding fluid environment. The tidal velocity is modeled as a sinusoidal function using predefined amplitude and frequency to replicate natural tidal cycles.

The core of the simulation uses MATLAB's ODE45 solver to numerically solve the second-order differential equation. This equation is converted into a system of first-order equations and implemented through a user-defined function. The solver computes the angular displacement and angular velocity of the turbine shaft over the specified time period.

Once the dynamic response is obtained, the code calculates the instantaneous tidal velocity at each time step and uses it to determine the hydrodynamic torque acting on the turbine blades. The power output is then computed as the product of torque and angular velocity, providing a time-dependent profile of power generation.

In addition to the physics-based simulation, a machine learning model is incorporated using polynomial regression. The simulated data is used to train the model, which establishes a relationship between tidal velocity and power output. This allows for quick prediction of power without solving the differential equations repeatedly.

Finally, the code generates plots to visualize angular displacement, angular velocity, and power output along with the predicted values from the machine learning model. These visualizations help in understanding the system behaviour and validating the accuracy of the predictive model.

Overall, the MATLAB implementation provides an efficient framework to analyse turbine dynamics, evaluate power generation, and integrate data-driven prediction techniques.

```matlab
clc;
clear;
close all;

%% PARAMETERS
rho = 1025;          % Seawater density (kg/m^3)
R = 5;               % Turbine radius (m)
A = pi*R^2;          % Swept area (m^2)
Ct = 0.4;            % Torque coefficient
I = 5000;            % Moment of inertia (kg.m^2)
c = 2000;            % Damping coefficient
k = 10000;           % Shaft stiffness

V0 = 2;              % Max tidal velocity (m/s)
omega = 0.2;         % Tidal frequency (rad/s)

tspan = [0 100];     % Time span

%% INITIAL CONDITIONS
y0 = [0; 0];         % [theta, theta_dot]

%% SOLVE DIFFERENTIAL EQUATION
[t, y] = ode45(@(t,y) turbine_dynamics(t,y,rho,A,R,Ct,I,c,k,V0,omega), tspan, y0);

theta = y(:,1);
theta_dot = y(:,2);

%% TIDAL VELOCITY
v = V0 * sin(omega * t);

%% TORQUE
Torque = 0.5 * rho .* A .* R .* Ct .* (v.^2);

%% POWER
Power = Torque .* theta_dot;

%% ================= ML MODEL (IMPROVED) =================
% Using both velocity (v) and angular velocity (theta_dot)

X = [v, theta_dot];              % Input features
Y = Power;                       % Output

% Add bias term
X_reg = [ones(length(X),1), X];

% Regression coefficients
coeff = X_reg \ Y;
```

```
% Prediction
Power_pred = X_reg * coeff;

%% DISPLAY MODEL
disp('ML Model: P = a0 + a1*v + a2*theta_dot');
disp(coeff);

%% PLOTS
figure;

subplot(3,1,1);
plot(t, theta);
title('Angular Displacement');
xlabel('Time (s)');
ylabel('\theta (rad)');

subplot(3,1,2);
plot(t, theta_dot);
title('Angular Velocity');
xlabel('Time (s)');
ylabel('\theta dot (rad/s)');

subplot(3,1,3);
plot(t, Power, 'b', 'LineWidth', 1.5); hold on;
plot(t, Power_pred, 'r--', 'LineWidth', 1.5);
title('Power Output & Improved ML Prediction');
xlabel('Time (s)');
ylabel('Power (W)');
legend('Actual','Predicted');
grid on;

%% FUNCTION
function dydt = turbine_dynamics(t,y,rho,A,R,Ct,I,c,k,V0,omega)

    theta = y(1);
    theta_dot = y(2);

    v = V0 * sin(omega * t);

    Torque = 0.5 * rho * A * R * Ct * v^2;

    theta_ddot = (Torque - c*theta_dot - k*theta) / I;

    dydt = [theta_dot; theta_ddot];
end
```

Figure 1 MATLAB Code

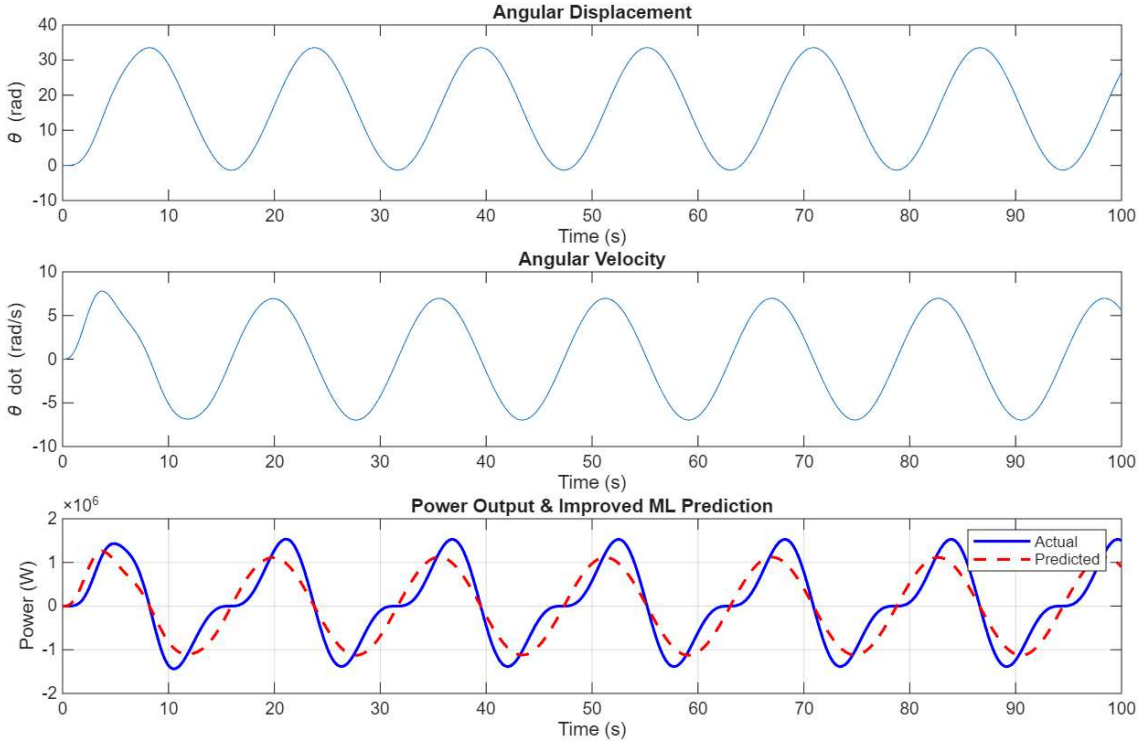

Figure 2 MATLAB Code Result

Limitations, Future Scope, and Alternate Approach

The current model is simplified and does not capture all real-world complexities. It ignores turbulence, variable torque coefficients, and electrical conversion efficiency. The machine learning model is also basic and may not generalize well to unseen conditions.

Future work can include advanced models such as computational fluid dynamics (CFD), real tidal data integration, and the use of artificial neural networks for better prediction accuracy. Alternate approaches may involve blade pitch control, adaptive systems, or hybrid renewable setups combining tidal and wind energy.

Conclusion

This project demonstrates how a tidal turbine system can be effectively modelled using nonlinear dynamics and simulated in MATLAB. The integration of machine learning provides an additional layer of efficiency by enabling quick power predictions.

The study highlights the potential of tidal energy as a reliable and sustainable power source while also addressing the challenges involved in its implementation. With further advancements in modelling and technology, tidal energy systems can play a significant role in the future energy landscape.